

CSES Problem Set

Planets and Kingdoms

[TASK](#) | [SUBMIT](#) | [RESULTS](#) | [STATISTICS](#) | [TESTS](#)**Time limit:** 1.00 s **Memory limit:** 512 MB

A game has n planets, connected by m teleporters. Two planets a and b belong to the same kingdom exactly when there is a route both from a to b and from b to a . Your task is to determine for each planet its kingdom.

Input

The first input line has two integers n and m : the number of planets and teleporters. The planets are numbered $1, 2, \dots, n$.

After this, there are m lines describing the teleporters. Each line has two integers a and b : you can travel from planet a to planet b through a teleporter.

Output

First print an integer k : the number of kingdoms. After this, print for each planet a kingdom label between 1 and k . You can print any valid solution.

Constraints

- $1 \leq n \leq 10^5$
- $1 \leq m \leq 2 \cdot 10^5$
- $1 \leq a, b \leq n$

Example

Input:

```
5 6
1 2
2 3
3 1
3 4
4 5
5 4
```

Output:

```
2
1 1 1 2 2
```